KERALA SSLC | BIOLOGY

CHEMORECEPTION IN ORGANISMS

PYQ • 2021 - 2026

1. 2026 **1 MARK**

Which hormone maintains dormancy of seeds until favourable conditions ?

(a) Auxin (b) Gibberellin (c) Ethylene (d) Abscisic acid

2. 2026 **3 MARKS**

Analyse the illustration given below and answer the following questions.

(X) → Signal → Flowering

(a) Identify the pigment labelled as 'X'.

(b) Which part of the plant produces this pigment ?

(c) How does 'X' induce flowering ?

3. 2026 **3 MARKS**

Observe the given blood test result of a person.

Fasting blood glucose (FBS) : 90 mg/dL HbA1c : 6.7 %

(a) Is the fasting blood glucose value normal ?

(b) Is the given HbA1c value normal ? Why ?

(c) Suggest any two life style habits that people should follow to maintain a normal blood glucose level.

4. 2025 **1 MARK**

Which is the main endocrine gland that controls metabolic activities ?

(a) Pancreas (b) Thyroid gland (c) Pineal gland (d) Adrenal gland

5. 2025 **3 MARKS**

The normal level of blood glucose is 70 – 110 mg/100 ml.

(a) What are the two reasons that cause glucose to rise above this level and be excreted in urine ?

(b) Mention one function of insulin and one function of glucagon in regulating blood glucose levels.

6. 2025 **4 MARKS**

Arrange columns B and C according to column A.

A – Gland : Pituitary, Adrenal gland, Ovary, Hypothalamus, (blank)

B – Hormone : Progesterone, Releasing hormone, Prolactin, Cortisol, Calcitonin

C – Function : Slows down the action of defense cells ; Production of milk ; Controls secondary sexual characters ; Implantation of embryo in the uterus ; Controls the production of tropic hormones

7. 2024 **1 MARK**

Identify the relationship in A and fill in B.

(A) Dormancy of embryo : Abscisic acid

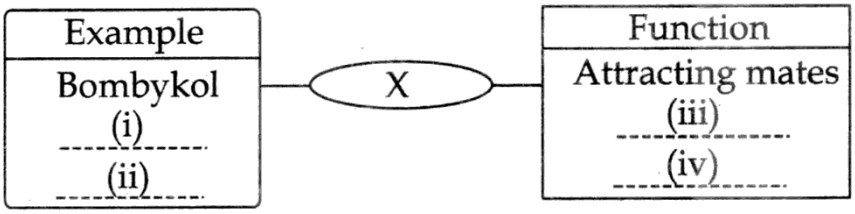
(B) Fruit formation : _____

8. 2024 **3 MARKS**

Analyse the illustration and answer the questions :

(a) Which chemical is indicated as X ?

(b) Fill in (i), (ii), (iii) and (iv).



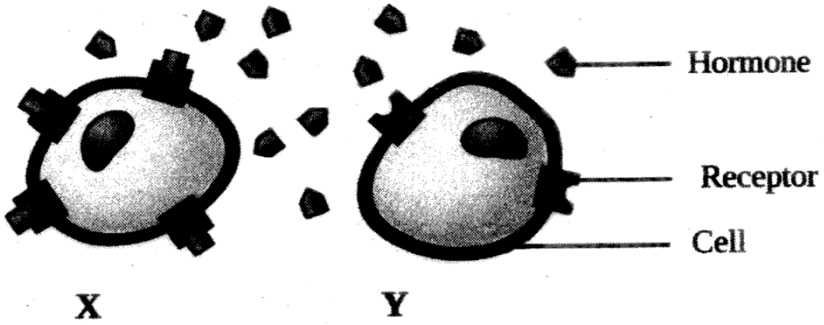
9. 2024 **4 MARKS**

Observe the illustration showing hormone action in cells X and Y and answer the questions.

(a) Identify the target cell of the hormone indicated. Why ?

(b) How do hormones change cellular activities ?

(c) What is the role of the hypothalamus in regulating the production of hormones ?



10. 2023 **1 MARK**

Correct mistake if any in the underlined part of the given statements.

(a) Gibberellin stimulates break down of stored food to facilitate germination.

(b) Ethylene promotes the growth of terminal buds.

(c) Cytokinin maintains the dormancy of embryo.

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11. 2023 **3 MARKS**

Table showing the blood glucose level of three persons before breakfast :

A : 82 mg/100 ml B : 178 mg/100 ml C : 104 mg/100 ml

(a) Which person is Diabetic ?

(b) What is the cause of this disease ?

(c) What are the symptoms of this disease ?

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12. 2023 **4 MARKS**

Analyse the illustration of hormonal disorders and answer the questions.

Over production of Hormone (X) : during growth phase → A ; after growth phase → B

Lower production of Hormone (X) : during growth phase → Dwarfism

(a) Name the hormone indicated as X.

(b) Identify the disorders A and B.

(c) What are the symptoms of A and B ?

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13. 2022 **1 MARK**

Choose the correct pair.

(a) Dormancy of embryo : Cytokinin

(b) Sprouting of leaves : Abscisic acid

(c) Fruit formation : Gibberellin

(d) Ripening of leaves and fruits : Ethylene

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14. 2022 **3 MARKS**

Analyse the statement and answer the questions.

'Ants moving in a line along a particular trail by the production of certain chemical substances'.

(a) Such chemicals are known as _____ .

(b) Write two examples for these chemical substances.

(c) Write other two uses of these chemicals substances.

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15. 2022 **5 MARKS**

Analyse the process related to the regulation of blood glucose level and answer the questions.

Glucose →(X)→ Glycogen →(Y)→ Glucose

(a) Identify the hormones indicated as X and Y.

(b) Name the gland which produces these hormones.

(c) Name the disease caused due to the reduced production of the hormone X.

(d) Write the symptoms of this disease.

(e) Write another action of the hormone Y in regulating the level of glucose in blood.

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16. 2021 **1 MARK**

Choose the correct pair.

(a) Prolactin : Production of milk (b) Aldosterone : Promotes growth

(c) Cortisol : Salt-water balance (d) Melatonin : Maintains blood pressure

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17. 2021 **1 MARK**

Which is known as youth hormone ?

(a) Adrenaline (b) Thymosin (c) Aldosterone (d) Testosterone

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18. 2021 **2 MARKS**

Complete the following table appropriately.

Symptoms (i) ; Cause : Production of dopamine in the brain gets reduced ; Disease (iii)

Symptoms : Loss of memory and inability to do routine works ; Cause (ii) ; Disease (iv)

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19. 2021 **2 MARKS**

Complete Column B of given table using suitable functions given in the box.

A – Plant hormones : (a) Gibberellin (b) Auxin (c) Abscisic acid (d) Abscisic acid and Ethylene

Box : Dropping of leaves and fruits ; Sprouting of leaves ; Fruit formation ; Dormancy of Embryo

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20. 2021 **2 MARKS**

Complete the table including the name of endocrine glands and their hormones responsible for the balance of Calcium level in blood.

Calcium level in blood : (a) increases – hormone (i), gland (ii)

(b) decreases – hormone (iii), gland (iv)

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21. 2021 **3 MARKS**

Analyse the given table and arrange Column B and C according to Column A.

A : Acromegaly, Cretinism, Gigantism

B : Increased production of somatotropin during growth phase ; Excessive production of somatotropin after growth phase ; Decreased production of thyroxine during infancy ; Excessive production of thyroxine

C : Overgrowth on the neck ; Excessive growth of the body ; Overgrowth of bones on the face, jaws and fingers ; Physical and mental growth retardation in children

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22. 2021 **4 MARKS**

Observe the illustration and answer the questions.

(a) Identify (i) and (ii).

(b) Write two functions of (ii).

(c) Name the condition resulted by the deficiency of (ii).

